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ORIGINAL ARTICLE

Artificial intelligence for stored product insect detection and control with laser beams: the Insect Laser project

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Abstract

From September 2017 to March 2021, the Insect Laser project was carried out. Three partners from stored product protection, micro-integration and laser research, as well as media technology and artificial intelligence (AI) worked on the task to provide image recognition for adult granary weevils (*Sitophilus granarius*; Coleoptera, Curculionidae) and adult Indianmeal moths (*Plodia interpunctella*; Lepidoptera, Pyralidae). Deep learning was facilitated by training the system with photos and short video clips. After training, the system was able to identify both granary weevils and Indianmeal moths with a confidence level of 93%. The camera for image recognition was mounted on a one-dimensional track equipped with an infrared LED lamp and a laser source at a distance of 500 mm to insects on the floor of the test chamber. Digital image processing was carried out with YOLO v4-Tiny. Location coordinates of an insect detected in the scanned camera frame were forwarded to the laser beamer unit. Both, adult granary weevils and Indianmeal moths, were controlled with beams at wavelengths of 808 nm (8 W) or 1470 nm (2.5 W) in treatment times of 500 ms. The results suggest that using the system in a large warehouse environment is less feasible. However, its use in a closed trap could provide workers safety and be effective for both monitoring and control.

Keywords Artificial intelligence · Deep learning · Image recognition · Position data transfer · Laser insect control

Introduction

Today, artificial intelligence (AI) is used for image recognition in many areas of daily life. Also in applied entomology, there is a growing body of publications dealing with this topic (Kariyanna and Sowjanya 2024; Meghraj and Choudhary 2024; Chacrabarty et al. 2025; Goswamiy 2025). This technology can also be useful in a storage environment, e.g. for monitoring stored product insects (Shen et al. 2013; Shi et al. 2020; Badgujar et al. 2023; Sun et al. 2026) because various minute species may infest certain products, and insect taxonomy is a difficult task for storage personnel. A good knowledge on the prevailing pest species can be important to make proper decisions for pest management.

Stored product insects may occur in a given premise for three different reasons (Stein 1986; Adler et al. 2021b):

1. The insects came with infested raw materials, packaging or palettes.
2. The insects have been present for some time and developed in residual products accumulated in unattended areas of the building, false ceilings and other crevices or within the machinery (Dowdy and McGaughey 1998).
3. The insects found a way to enter from outside, attracted by favourable temperature, moisture and/or attractive volatiles (Hagstrum 2001; Anukiruthika and Jayas 2025).

Infestation pressure in moderate climates is expected to rise with global warming as stored product insects are of tropical origin and prefer higher temperature and moisture contents. In species already present higher annual temperatures may lead to faster reproduction and more generations per

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year. Moreover, Mediterranean species seem to move further north into temperate climates (Adler et al. 2022).

The adult stage of stored product pest insects is not only the reproductive stage but may also cover larger distances to find new sites for feeding, mating and oviposition. Suitable sites attract insects by volatile cues emitted either by the stored goods or by volatiles such as pheromones of conspecifics and kairomones of other species that signal a certain level of infestation. In order to find suitable substrates, stored product insects can recognize odoriferous substances emitted from attractive stored products over long and short distances (Adhikary et al. 2016; Ndomo et al. 2014, 2015). This orientation via volatile cues allows insects and other arthropods to migrate purposefully via openings into buildings. Stored product insects may even find minute openings to enter leaky packaging (Olsson 2001; Adler 2004; Olsson et al. 2006; Uechi et al. 2007; Böye et al. 2014). On short distance, stored product insects are also capable of detecting areas with higher moisture by specialized sensilla on their antenna (Loschiavo 1983; Enjin 2017). To summarize the previous findings and observations, the first individuals that appear at a new or yet un-infested storage site are prone to be adults.

Artificial intelligence can be trained for image recognition comparing learnt images with new images recorded by a surveillance camera. An early detection and species identification can help to maintain product quality if prompt action follows. A combination of early detection and physical control measures could potentially prevent a severe infestation at an early stage. At the same time, it could reduce treatment costs and the dependence on synthetic pesticides.

The idea of the InsectLaser project described here was to train an AI to detect adult granary weevils *Sitophilus granarius* (Col., Curculionidae) and adult Indianmeal moths *Plodia interpunctella* (Lep., Pyralidae) as representatives of distinctly different stored product pests. Once detected, the location of the found specimen was to be transferred to a laser beamer which then would rapidly emit a high-energy laser beam to control the given insect before the latter was able to move away from the recorded position.

Up until now, laser control has been mainly used elsewhere in Integrated Pest Management:

A study by Andreasen et al. (2022) described laser weed control with autonomous field rovers. A similar, more recent study (Sosnoskie et al. 2025) demonstrated that laser weeding had significant advantages to using herbicides. Both studies followed a similar approach as the project described here, the automatic optical detection of harmful organisms and their control by laser, which underlines the general interest in this technology.

A different project in applied entomology developed a laser system called “photonic fence” to identify, track, and shoot down small flying mosquitoes in the wild in order to reduce the population of *Anopheles stephensi* (Diptera, Culicidae) as a vector of malaria disease (Keller 2016; Rakhmatulin 2021). Another study compared identification, tracking, and control of the same species with that of a psyllid, *Diaphorina citri* (Mullen et al. 2016). Other authors evaluated the use of laser beams to control mainly the nymphs of aphids in the field (Gaetani et al. 2021). And yet another group of researchers studie the control of the whitefly Bemisia tabaci by laser beams (Zaidem et al. 2023).

In the project described here, a trained camera-AI system was to scan surfaces in periodic intervals. As soon as an insect was detected, image recognition decided if this is a target insect. If the decision was positive, location coordinates were forwarded to a laser beamer that emitted a beam, directed by a parabolic mirror to control the respective insect. To our knowledge so far, no research project used AI image recognition and laser beams for pest control in the context of stored product protection. We intended to find answers to the following main research questions:

- Is it possible to train AI to recognize images of live granary weevils and Indianmeal moths in order to detect these species with a high reliability?
- Can a moving camera scan surfaces and compare picture frames to stored data, decide for hits and transfer location data to a laser beamer within some fractions of a second?
- Is there a laser that emits beams strong enough to kill our target insects instantly but without harming stored grain surfaces below the target insect? As a combination of wave length and energy define the depth of action of a laser beam, the resulting effects were to be studied here.

There were concerns, whether the transmission of position data to a laser emitter would be fast enough and whether a laser beam could reliably kill insects in a short time without damaging or blackening grains or surfaces. This led the project-funding unit (BLE) to divide funding into two equal phases and to link the start of the second phase to the achievement of the aforementioned milestones (see research questions).

Materials and methods

Test insects

The target insects selected were the crawling, flightless granary weevil (*Sitophilus granarius*; Coleoptera, Curculionidae) and the flying Indianmeal moth (*Plodia interpunctella*, Lepidoptera, Pyralidae). Both species were chosen as

representatives for stored product insects because the granary weevil is a major grain pest, flightless with reduced elytra and a reduced second pair of wings, has a hard outer skeleton (cuticle) and can be found crawling on surfaces. In contrast, the Indianmeal moth, a small flying lepidopteran of the pyralid family, is a major pest in food industry with a long list of stored products suitable for attack such as dried fruits, teas and spices, nuts and chocolates, as well as grain and grain products. This species is rather delicate with a soft body, prefers to rest on walls or other vertical surfaces, and visits the flat surface of cereal grains and other products mainly for oviposition. As in other butterflies, adult moths do not feed on hard products but are rather short-lived and have sucking mouthparts to pick up liquids only. In stored product moths, the larval stages cause the main damage by feeding, by their excretions and by producing extensive webbings.

The Indianmeal moth was cultivated on wheat bran fortified with some yeast, glucose and glycogen and broken almonds at 25 ± 1 °C and 65–70% rh. Adult moths were kept on an oviposition sieve container at similar conditions for three days. The eggs, deposited through the sieve were cleaned and approximately 500 eggs were used for a new culture jar. The granary weevil was cultivated on whole wheat grains at similar climatic conditions. Some 2500 adults of both sexes were placed onto 142 g of wheat grains with a moisture content of 14 ± 0.5% for a period of three days and sieved off thereafter. This procedure was repeated for both insects every week. At these environmental conditions, both species took about six weeks for the development from eggs to adults and have been kept at JKI for several years.

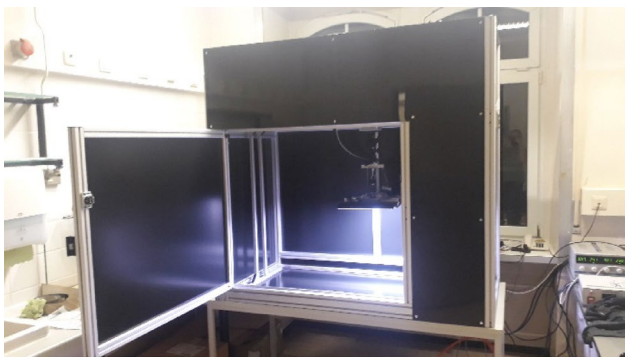


Fig. 1 View into laser protective housing box with safety-door, horizontally mobile LED lamp, and camera fixed in the center of the lamp. The laser beamer was mounted onto the rim of the LED lamp. While for insect detection infrared rays were used, the lamp emits visible light in this picture. For testing, insects were released on a grain or other surface at the bottom of the box at 500 mm distance to the camera (Photo: P. Kern, JKI)

Insect arena

For safety, a laser protective housing box was assembled from parts provided by ITEM Industrietechnik GmbH (item24.com, Solingen, Germany). According to workers safety regulations for the type 4 laser used, the door to the laser box was equipped with a lock that cut electricity (and thus laser beams) when opened. This is required for laser rays that may damage the human eye. Sheets of the laser protective housing box consisted of 3 mm thick black composite panels with polyethylene core material.

Laser protective housing box (experimental chamber) (Fig. 1).

Cameras, lamps, laser and image recognition

Cameras used in this study were a Kamera HXC40NIR, a Kamera HXC40c, and the camera link frame grabber PCIe-CL microEnable IV VD4 -CL (Baumer.com, Baumer GmbH, Friedberg, Germany). The cameras recorded 30 frames per second (FPS) and a frame of 2048 × 2048 pixles. The lamp was a LED light source (RGB) with a central gap for the camera, provided by Falcon (falconeyeshk.com, Hongkong, China). In addition, an infrared (iR) LED lamp and a Smartek Strobe Controller HPSC1-E from Framos GmbH (framos.com, Taufkirchen, Germany) were part of the equipment.

Within the first 18 months of the project, an image recognition system was selected. The camera, sensitive in the infrared range, had a resolution of 2048 × 2048 pixels and the iR source was in the wavelength range at 850 nm. This wavelength was chosen because insects have no vision in the iR range and are thus not disturbed by the monitoring rays nor the control rays (see below). The AI system was to be trained with image data of both species to detect grain weevils or Indianmeal moths with a high degree of confidence and to distinguish them from a smooth background or wheat grain surface. In image recognition, substrates such as grains compared to a smooth surface like paper may also play a role. Thus, the AI system had to learn distinguish an insect from gaps and shadows between grains during a process of machine learning (deep learning).

To speed up the results of the training data, 16 sub-frames of 512 × 512 pixels were generated from one camera frame. This size of frame proved an optimal section for image recognition in relation to the insect size and data processing. We discussed pulsed laser beams in order to reduce heating or burning effects on surfaces. The mobile positioning system Festo GmbH (festo.com, Esslingen, Germany) fixed camera and lamp at the ceiling of the laser box and allowed bi-directional movement on a single track.

Key components of the laser unit were the actual lasers (an 808 nm wavelength fiber coupled diode laser by DILAS, Germany, and an alternative 1470 nm diode laser fiber coupled by IZM), a beam forming and static deflection unit (a parabolic mirror mounted in an optomechanical cage system supplied by Thorlabs), as well as an industrial galvo scanner (Raylase, Germany). Both laser wavelengths, 808 nm and 1470 nm, were used to test control of fixed adult grain weevils and dried fruit moths, as well as their effects on grain, wood, and metal. A PT1000 temperature sensor was placed at the surface treated with laser beams in order to monitor the resulting temperature changes.

Results

With the chosen system it was possible to detect both adult granary weevils and adult Indianmeal moth and to determine both species accurately. Using a neural network, a recognition rate of 93% within 40 ms was achieved from approximately 94,000 data sets. The process of image recognition as depicted by blue and red circles around moths and weevils can be seen in Fig. 2. The position data of detected insects were transferred to a laser beamer within some 50 ms. Rapid transfer of position data to a laser source was possible by changing from digital image processing software Matlab to You-only-look-once-(YOLOv4-Tiny) open-source software, known from autonomous driving.

Proper aiming on the target insect was tested with a pilot laser emitting infrared light and recorded by an infrared camera (Fig. 3). In addition, the laser for physical insect control had to be able to emit a beam sufficient to kill the target insects while not leading to burns or blackening in the grain or on other surfaces.

With the 808 nm laser, we observed a more external effect. At 8.44 W optical power, the moth wings were damaged, shrinkage and blackening of the wings occurred. A slight discoloration effect of the carapace was visible in granary weevils. All tested individuals were killed. Both insect species produced an odour but no smoke.

Control of both species took place without any visible damage to the grain kernels below. The grain did not show any damage even at higher optical power levels of 13 W. Only when applying significantly higher power levels of 33 W, a slight blackening was visible. Wood behaved more critically than grain, but here, too, no damage occurred at

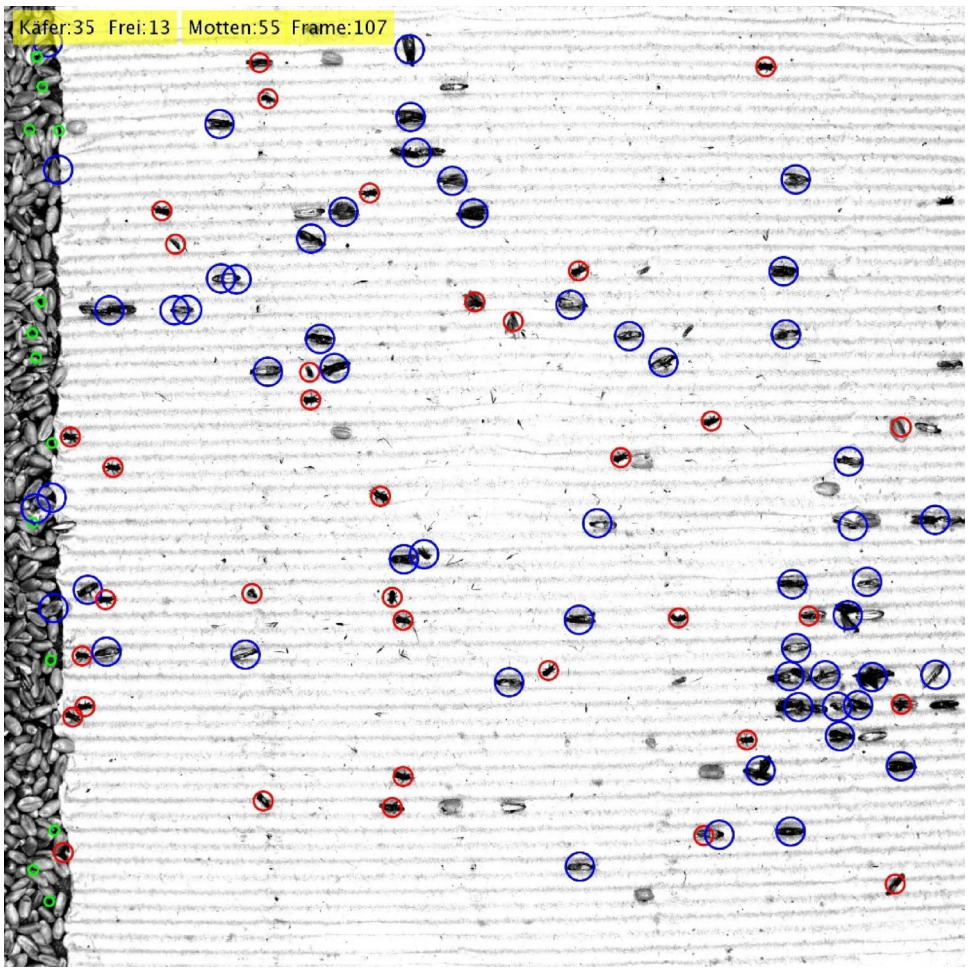


Fig. 2 Image recognition by the camera in combination with AI. The system had been trained with pictures and video clips of the respective species. Red and blue circles show individual moths or weevils recognized by the system, green circles empty voids between grain kernels (Photo: K. Große, BTU)

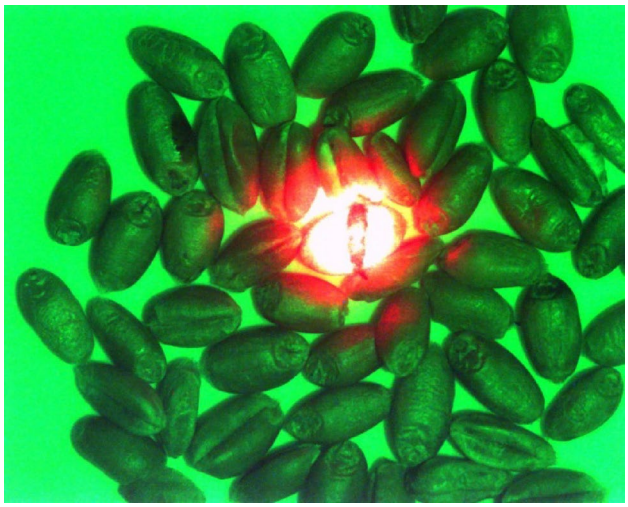


Fig. 3 A harmless pilot laser with infrared rays (635 nm) illuminating the targeted insect as recorded by an infrared camera. The effective laser beam with 2–2.5 W at 808–1470 nm would outshine the image (Photo: G. Böttger, ZMI)

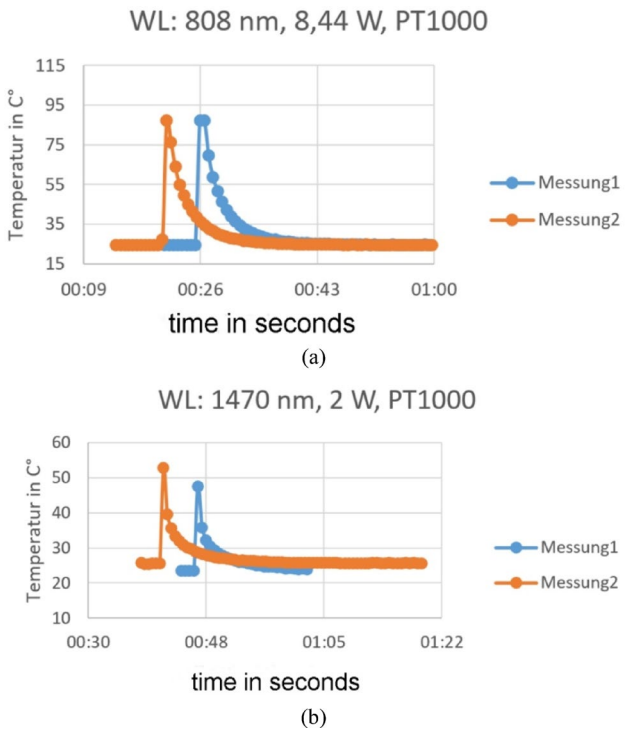


Fig. 4 **a**, **b** Surface temperatures caused by laser beams at wave lengths of 808 µm and electric power consumption of 7.44 W vs. 1400 µm and 2 W, as measured with a PT 1000 temperature sensor (Photo: G. Böttger, ZMI)

the required effective laser power (at 808 nm approx. 8 W and 500 ms, at 1470 nm approx. 2.5 W and 500–2000 ms). Surface temperatures as measured with a PT1000 sensor at 808 nm and 8.44 W optical power reached 90 °C after 500 ms while at 1470 nm and 2 W a maximum of only 55 °C was recorded (Fig. 4).

Discussion

Controlling adult insects on a grain surface would not have damaged grain kernels but would have left behind dead individuals as a potential contaminant. However, these adults would not have been available for reproduction. And if females were controlled prior to oviposition, an infestation could have been prevented. Cleaning out dead individuals can be achieved by normal grain cleaning as carried out in mills.

The laser with a wavelength of 1470 nm is a standard laser used in different applications (such as ablation in medical surgery or optical industrial processes) and is thus mass-produced at lower costs. As it gave better results at lower energy input, this could render a practical application more economical.

The 1470 nm laser had a complementary effect obviously more on the soft tissue in the body and was already fully effective at 2 W power and 500 ms. This is attributed to better absorption of this wavelength in watery matter. Ubaid (2016) had tested visible lasers at 405 nm and 110 W for periods between 10 min and 40 min against adult beetles and larvae of *Tribolium castaneum* and *Trogoderma granarium*. While 40 min exposure time gave up to 100% mortality in adult *T. castaneum* and 66.6% mortality in *T. granarium* this shows that visible lasers are much less effective than infrared lasers, the latter requiring much less energy at a shorter exposure time for a similar effect.

At the end of the project, a suspension for horizontal and one-dimensional camera movement, with a speed of 20 mm/s and a working distance of 500 mm for image recognition was required. Infrared rays proved sufficient for camera detection and did not affect insect behaviour.

Because of this working distance required for image recognition in order to maintain the pixel numbers at the pretrained level, the camera could not be turned to the wall in order to scan for insects in corners or on vertical surfaces as such a movement would have modified the distance between insect and camera. The implementation of a zoom and autofocus may be possible to allow for a sufficient resolution from a greater distance. However, this was technically not feasible in the project described here. This fact currently limits the commercial use of our results. Direct commercial exploitation of the overall system is not planned at present, as a practical prototype has yet to be developed. However, a trap system equipped with AI for infestation monitoring may be feasible even with the rather short distance tested. Keeping the laser unit inside the housing of a trap would add to workers safety and would keep dead insects separate from surfaces and stored products in food processing. Phillips and Throne (2010) mention “Processing facilities and

warehouses usually rely on trap captures for decision-making, a process that needs further research to optimize". Thus,

Automated pest monitoring using AI is a process developed now and may soon help farmers and food industry to react faster and to make better informed decisions. Especially the combination with a physical means of control like laser beams appears feasible. In a modern food factory with hermetically sealed structures and good sanitation standards, adult insects may be the occasional indruder. Here a trap that attracts, identifies, counts, and controls stored product insects and forwards the data to quality management could be a step forward.

Considering recent crises in world economy, global warming, pandemics, energy conflicts, and war, disrupting international food trade, food safety has once again become an issue on the public agenda. Besides, research in stored product protection and the combination with optic surveillance and AI remains an exciting topic.

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Declarations

Conflict of interest All authors declare that they have no competing interests that could have influenced the data.

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